

# Algebraic Geometry Problems

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## 1 Chapter 1

### 1.1.A:

- (a) Make sense of: “A group is a groupoid with one object”.

**Answer.**

Note that morphism composition is associative, that  $id_G \in \text{Mor}(G, G)$ , and that  $\text{Mor}(G, G)$  is closed. Since  $G$ , (as a category) is a groupoid, every morphism in  $\text{Mor}(G, G)$  is an isomorphism. Therefore for every  $\phi \in \text{Mor}(G, G)$ ,  $\exists! \phi^{-1} \in \text{Mor}(G, G)$ , so that

$$\phi \circ \phi^{-1} = \phi^{-1} \circ \phi = id_G$$

Then the set  $\text{Mor}(G, G)$  with the operation

$$\circ : \text{Mor}(G, G) \times \text{Mor}(G, G) \rightarrow \text{Mor}(G, G)$$

admits group structure. (Dropping the condition that  $G$  is a groupoid gives  $\text{Mor}(G, G)$  the structure of a monoid.)

- (b) Describe a groupoid that is not a group.

**Answer.**

Let  $\mathcal{C}$  be a groupoid with two objects  $A, B$ . Note that in a group, composition of any two elements is defined, however in our category  $\mathcal{C}$ , we have things like  $\phi, \psi \in \text{Mor}(A, B)$  which you can not compose because the sources and targets do not match.

### 1.1.B

Let  $A \in \text{Obj}(\mathcal{C})$ . Show that the invertible elements of  $\text{Mor}(A, A)$  form a group. (Called the automorphism group  $\text{Aut}(A)$ ). What are the automorphism groups for **Sets** and **Vec<sub>k</sub>**? Prove that isomorphic objects have isomorphic automorphism groups.

**Answer.**

Let  $A \in \text{Obj}(\mathcal{C})$  and consider the subset  $\text{Aut}(A) \subseteq \text{Mor}(A, A)$  with the property  $f \in \text{Aut}(A) \iff f$  is invertible. Then notes  $\text{id}_A \in \text{Aut}(A)$ , which serves as the identity of the group. As before, morphism composition is associative, and  $\forall f \in \text{Aut}(A)$ , since  $f$  is invertible,  $\exists f^{-1} \in \text{Aut}(A)$  such that  $f \circ f^{-1} = f^{-1} \circ f = \text{id}_A$ . Therefore,  $\text{Aut}(A)$  satisfies the group axioms.

If  $X \in \text{Obj}(\mathbf{Sets})$ ,  $\text{Aut}(X)$  consists of permutations of the elements of  $X$ . These are functions  $f : X \rightarrow X$  which are invertible (bijections). If  $V \in \text{Obj}(\mathbf{Vec}_k)$  is a vector space,  $\text{Aut}(V)$  consists of invertible matrices with entries from  $k$ .

*Proof.* Let  $A$  and  $B$  be isomorphic objects, so  $\exists f : A \rightarrow B$  and  $g : B \rightarrow A$  such that  $g \circ f = \text{id}_A$  and  $f \circ g = \text{id}_B$ . We know  $\text{Aut}(A)$  and  $\text{Aut}(B)$  are groups. Define the maps

$$\pi : \text{Aut}(A) \rightarrow \text{Aut}(B)$$

by

$$\pi : \phi \mapsto f \circ \phi$$

and

$$\pi^{-1} : \text{Aut}(B) \rightarrow \text{Aut}(A)$$

$$\pi^{-1} : \psi \mapsto g \circ \psi$$

and then

$$\pi^{-1}(\pi(\phi)) = g \circ (f \circ \phi) = (g \circ f) \circ \phi = \text{id}_A \circ \phi = \phi$$

$$\pi(\pi^{-1}(\psi)) = f \circ (g \circ \psi) = (f \circ g) \circ \psi = \text{id}_B \circ \psi = \psi$$

So  $\pi^{-1} \circ \pi = \text{id}_{\text{Aut}(A)}$  and  $\pi \circ \pi^{-1} = \text{id}_{\text{Aut}(B)}$ . Therefore  $\text{Aut}(A) \cong \text{Aut}(B)$ .  $\square$

### 1.2.A:

**Definition 1.** An object in a category  $\mathcal{C}$  is initial if it has exactly 1 map to each object in  $\mathcal{C}$ . An object is terminal (or final) if it has exactly one map from each object. An object is a zero object if it is both initial and terminal.

Show that any two initial objects are uniquely isomorphic. Same for terminal.

*Proof.* Let  $A, A'$  be two initial objects. Then, since  $A$  is initial,  $\exists! f : A \rightarrow A'$ . Similarly, since  $A'$  is initial,  $\exists! g : A' \rightarrow A$ . Then  $f \circ g : A' \rightarrow A'$  is unique, but  $id_{A'} : A' \rightarrow A'$  is a map  $A' \rightarrow A'$ . So  $f \circ g = id_{A'}$ . Similarly,  $g \circ f : A \rightarrow A$  is unique, so  $g \circ f = id_A$ . Therefore,  $A \cong A'$  uniquely. For terminal, flip the arrows.  $\square$

### 1.2.B:

What are initial, terminal, and zero objects in **Sets**, **Rings**, **Top**?

**Answer.**

- In **Sets**, the empty set  $\emptyset$  maps uniquely into every set  $X$ , via the empty function (which acts as the inclusion  $\emptyset \hookrightarrow X$ ), so it is initial. However, you can not define a function  $\phi : X \rightarrow \emptyset$  for all  $X$ , so it is not a terminal object. For any singleton set  $\{a\}$ , there is a unique map  $g : X \rightarrow \{a\}$ ,  $g(x) = a$  for all  $x \in X$ . Therefore, each singleton set is terminal.
- In **Rings**, the zero ring  $\{0\}$  is terminal, because for any ring  $R$ , take  $f : R \rightarrow \{0\}$ ,  $f(r) = 0$  for all  $r \in R$ . This is a valid ring hom, and it is unique. The ring  $(\mathbb{Z}, +, \cdot)$  is initial, because for any ring  $R$ , we must have (by our conventions)

$$f : \mathbb{Z} \rightarrow R, f : 1_{\mathbb{Z}} \mapsto 1_R$$

and since  $\mathbb{Z}$  is generated by  $1_{\mathbb{Z}}$ , that rule determines the entire homomorphism. (Note: this map captures the Characteristic of the ring  $R$ , because it is the  $n$  in  $\ker f = n\mathbb{Z}$ ).

- In **Top**, similar to **Sets**, the empty space  $\emptyset$  (with its corresponding topology) is initial via the empty function, which is trivially continuous. The terminal objects are singletons  $\{a\}$ , because there is a unique map  $f : X \rightarrow \{a\}$ ,  $f(x) = a$ , which is continuous because the preimage of  $a$  is  $X$ , which is open.

### 1.2.C:

Show that  $A \rightarrow S^{-1}A$ ,  $(a \mapsto \frac{a}{1})$  is injective iff  $S$  contains no zerodivisors.

*Proof.* ( $\implies$ ) : Let  $f : A \rightarrow S^{-1}A$ ,  $f : a \mapsto \frac{a}{1}$  be injective. Let  $a \neq b$ , then  $f(a) \neq f(b) \implies \frac{a}{1} \neq \frac{b}{1}$ . That means

$$\nexists s \in S, \text{ so that } s(a - b) = 0$$

So for  $x = (a - b) \in A$  nonzero, there is no  $s \in S$  so that  $s \cdot x = 0$ . Therefore,  $S$  contains no zerodivisors.

( $\impliedby$ ) : Let  $S$  be a multiplicative set with no zerodivisors. Let  $f : A \rightarrow S^{-1}A$ ,  $f : a \mapsto \frac{a}{1}$  be a function. Let  $f(a) = f(b)$  for some  $a, b \in A$ . Then  $\frac{a}{1} = \frac{b}{1} \implies \exists s \in S$  such that  $s(a - b) = 0$ . Since  $s$  is not a zerodivisor, we must have  $a - b = 0 \implies a = b$ . Therefore  $f$  is injective.  $\square$

### 1.2.D:

Verify  $A \rightarrow S^{-1}A$  satisfies the following universal property:  $S^{-1}A$  is initial among  $A$ -algebras  $B$  where every element of  $S$  is sent to an invertible element in  $B$ . Translation: any map  $A \rightarrow B$  where every  $s \in S$  is sent to an invertible in  $B$  must factor uniquely through  $A \rightarrow S^{-1}A$ .

**Remark.** An  $A$ -module  $M$  is the same as a map

$$\rho : A \rightarrow \text{End}_{\mathbb{Z}}(M)$$

$$\rho : s \mapsto \mu_s$$

where

$$\mu_s : M \rightarrow M$$

$$\mu_s : m \mapsto s \cdot m$$

And to say that a scalar  $s$  is invertible is to say that  $\mu_s \in \text{End}_{\mathbb{Z}}(M)$  is a bijection (automorphism).

*Proof.* Let  $f : A \rightarrow S^{-1}A$  where  $f(a) = \frac{a}{1}$ . Let  $B$  be an  $A$ -algebra, with map  $\phi : A \rightarrow B$  such that  $\phi(s)$  is a unit in  $B$ . We claim  $\exists! g$  such that

$$\begin{array}{ccc} A & \xrightarrow{f} & S^{-1}A \\ & \searrow \phi & \downarrow !g \\ & & B \end{array}$$

commutes. Uniqueness is shown by the following:

$$g\left(\frac{a}{s}\right) = g(f(a) \cdot f(s)^{-1}) = g(f(a)) \cdot g(f(s))^{-1} = \phi(a) \cdot \phi(s)^{-1}$$

So what  $g$  does to  $\frac{a}{s} \in S^{-1}A$  is completely determined by  $\phi : A \rightarrow B$ . To show existence we need to show that  $g$  is well-defined and that it is a ring hom.

Well-defined: Let  $\frac{a'}{s'} = \frac{a}{s}$ . Then we want  $g\left(\frac{a'}{s'}\right) = g\left(\frac{a}{s}\right)$ .

$$\frac{a'}{s'} = \frac{a}{s} \implies \exists t \in S \text{ such that } t(a's - s'a) = 0$$

This is an expression in  $A$ , so apply  $\phi : A \rightarrow B$  to both sides:

$$\phi(t)(\phi(a's) - \phi(as')) = 0$$

Since  $t \in S$ ,  $\phi(t)$  is a unit, so it is not a zerodivisor. Therefore  $\phi(a's) = \phi(as') \implies \phi(a')\phi(s) = \phi(a)\phi(s')$ . Since  $\phi(s)$  and  $\phi(s')$  are units in  $B$ , we can multiply by  $\phi(s)^{-1}$  and  $\phi(s')^{-1}$  to get:

$$\phi(a')\phi(s')^{-1} = \phi(a)\phi(s)^{-1} \implies g\left(\frac{a'}{s'}\right) = g\left(\frac{a}{s}\right)$$

As required, so  $g$  is well-defined.

Ring homomorphism: We check  $g(1_{S^{-1}A}) = 1_B$ :

$$\begin{aligned} g\left(\frac{1}{1}\right) &= \phi(1_A)\phi(1_A)^{-1} \\ &= 1_B \cdot 1_B \\ &= 1_B \end{aligned}$$

$g$  respects addition:

$$\begin{aligned}
g\left(\frac{a}{s} + \frac{b}{t}\right) &= g\left(\frac{at + bs}{st}\right) \\
&= \phi(at + bs)\phi(st)^{-1} \\
&= [\phi(a)\phi(t) + \phi(b)\phi(s)]\phi(s)^{-1}\phi(t)^{-1} \\
&= \phi(a)\phi(s)^{-1} + \phi(b)\phi(t)^{-1} \\
&= g\left(\frac{a}{s}\right) + g\left(\frac{b}{t}\right)
\end{aligned}$$

$g$  respects multiplication:

$$\begin{aligned}
g\left(\frac{a}{s} \cdot \frac{b}{t}\right) &= g\left(\frac{ab}{st}\right) \\
&= \phi(ab)\phi(st)^{-1} \\
&= \phi(a)\phi(b)\phi(s)^{-1}\phi(t)^{-1} \\
&= \phi(a)\phi(s)^{-1}\phi(b)\phi(t)^{-1} \\
&= g\left(\frac{a}{s}\right) \cdot g\left(\frac{b}{t}\right)
\end{aligned}$$

So

$$\begin{aligned}
g : S^{-1}A &\rightarrow B \\
g : \frac{a}{s} &\mapsto \phi(a)\phi(s)^{-1}
\end{aligned}$$

is a well defined, unique ring homomorphism which makes the diagram commute. Therefore, every map  $\phi : A \rightarrow B$  (every  $A$ -algebra  $B$ ) factors uniquely through  $S^{-1}A$ .  $\square$

### Localization of modules, mini problem:

Let  $M$  be an  $A$ -module. Then we define  $\phi : M \rightarrow S^{-1}M$  as being initial among  $A$ -module maps  $M \rightarrow N$  such that elements of  $S$  are invertible in  $N$ . Or,  $M \rightarrow S^{-1}$  is universal among  $A$ -module maps from  $M$  to modules that are actually  $S^{-1}A$ -modules. Mini problem: Define the objects and morphisms in this category to make this precise.

**Answer.**

- The objects of this category are pairs  $(N_i, \psi_i)$  where  $N_i$  is an  $A$ -module and  $\psi : M \rightarrow N_i$  is an  $A$ -module homomorphism such that the multiplication map  $s \times - : N_i \rightarrow N_i$  is an automorphism.
- The morphisms of this category are: Let  $(N_1, \psi_1), (N_2, \psi_2)$  be two objects of the category. Then they both define maps from  $M$ :

$$\begin{array}{ccc} M & \xrightarrow{\psi_1} & N_1 \\ \psi_2 \downarrow & \swarrow \theta & \\ N_2 & & \end{array}$$

The map  $\theta : N_1 \rightarrow N_2$  of  $A$ -modules, such that the triangle commutes ( $\theta \circ \psi_1 = \psi_2$ ) is an element of  $\text{Mor}((N_1, \psi_1), (N_2, \psi_2))$ .

### 1.2.E:

Show that  $\phi : M \rightarrow S^{-1}M$  exists by constructing something satisfying the universal property. Hint: define elements of  $S^{-1}M$  to be  $\frac{m}{s}$  where  $m \in M$ ,  $s \in S$ , with equality given by

$$\frac{m_1}{s_1} = \frac{m_2}{s_2} \iff \exists t \in S \mid t(m_1 s_2 - m_2 s_1) = 0$$

Addition given by

$$\frac{m_1}{s_1} + \frac{m_2}{s_2} = \frac{m_1 s_2 + m_2 s_1}{s_1 s_2}$$

And the  $S^{-1}A$ -module structure (and therefore the  $A$ -module structure) is given by

$$\frac{a}{s} \cdot \frac{m_1}{t} = \frac{a \cdot m_1}{st}$$

**Remark.** The  $A$ -module structure is achieved by “pulling back along the localization map” like so: Given  $\psi : S^{-1}A \rightarrow \text{End}_{\mathbb{Z}}(S^{-1}M)$ , we take  $f : A \rightarrow S^{-1}A$  and compose to get  $\rho_A : A \rightarrow \text{End}_{\mathbb{Z}}(S^{-1}M)$  where  $\rho_A = \psi \circ f$

*Proof.* Let  $M$  and  $S^{-1}M$  be  $A$ -modules, with  $S^{-1}M$  defined as above. Define the map of  $A$ -modules

$$\begin{aligned}\phi : M &\rightarrow S^{-1}M \\ \phi : m &\mapsto \frac{m}{1}\end{aligned}$$

We want to show that this map exists and that it satisfies the universal property. The category we are working in is the (full) subcategory of the slice category  $M$ , where objects are pairs  $(N, \psi)$  where  $\psi : M \rightarrow N$  and every element of  $S$  maps to an invertible in  $N$ . Morphisms, say between  $(N_1, \psi_1)$  and  $(N_2, \psi_2)$  are maps  $\theta : N_1 \rightarrow N_2$  which make

$$\begin{array}{ccc} M & \xrightarrow{\psi_1} & N_1 \\ \psi_2 \downarrow & \swarrow \theta & \\ N_2 & & \end{array}$$

commute (so  $\psi_2 = \theta \circ \psi_1$ ). So we want to show  $\phi : M \rightarrow S^{-1}M$  is initial in this category (or, the pair  $(S^{-1}M, \phi)$  is an initial object).

So, let  $(N, \psi)$  be an object of this category. We want to show there  $\exists! \theta : S^{-1}M \rightarrow N$  such that

$$\begin{array}{ccc} M & \xrightarrow{\phi} & S^{-1}M \\ \psi \downarrow & \swarrow \theta & \\ N & & \end{array}$$

commutes ( $\psi = \theta \circ \phi$ ).

Uniqueness of  $\theta$  is seen by:

$$\begin{aligned}\theta\left(\frac{m}{s}\right) &= \theta\left(\mu_s^{-1}(\phi(m))\right) \\ &= \mu_s^{-1}(\theta(\phi(m))) \\ &= \mu_s^{-1}(\psi(m))\end{aligned}$$

Where the second equality is due to  $\theta(\mu_s(m)) = \mu_s(\theta(m))$ , then multiply by  $\mu_s^{-1}$  (which exists) on both sides.

Well-defined: Let  $\frac{m}{s} = \frac{m'}{s'}$ , then  $\exists t \in S$  with  $t(ms' - m's) = 0$ . Apply  $\psi : M \rightarrow N$  to both sides:

$$t(\psi(m)s' - \psi(m')s) = 0$$

Since  $t \in S$  is a unit, it is not a zerodivisor, so  $s' \cdot \psi(m) = s \cdot \psi(m')$ . Since  $s, s' \in S$ ,  $s \cdot s' \in S$  and so  $\mu_{s \cdot s'}^{-1}$  exists. Multiply by this on both sides:

$$\mu_s^{-1}\psi(m) = \mu_{s'}^{-1}\psi(m') \implies \theta\left(\frac{m}{s}\right) = \theta\left(\frac{m'}{s'}\right)$$

$\theta$  respects addition:

$$\begin{aligned} \theta\left(\frac{m_1}{s_1} + \frac{m_2}{s_2}\right) &= \theta\left(\frac{m_1s_2 + m_2s_1}{s_1s_2}\right) \\ &= \mu_{s_1s_2}^{-1}(\psi(m_1)\mu_{s_2} + \psi(m_2)\mu_{s_1}) \\ &= \mu_{s_1}^{-1}(\psi(m_1)) + \mu_{s_2}^{-1}(\psi(m_2)) \\ &= \theta\left(\frac{m_1}{s_1}\right) + \theta\left(\frac{m_2}{s_2}\right) \end{aligned}$$

$\theta$  respects  $A$ -scalar multiplication:

$$\begin{aligned} \theta\left(a \cdot \frac{m}{s}\right) &= \theta\left(\frac{a \cdot m}{s}\right) \\ &= \mu_s^{-1}(\psi(a \cdot m)) \\ &= \mu_s^{-1}(a \cdot \psi(m)) \\ &= a \cdot \mu_s^{-1}(\psi(m)) \\ &= a \cdot \theta\left(\frac{m}{s}\right) \end{aligned}$$

Where the 4th equality is due to  $\mu_s^{-1}$  being the inverse of an  $A$ -linear operation, so it is  $A$ -linear.

So  $\theta$  exists and is unique, therefore  $(S^{-1}M, \phi)$  is initial in this category.  $\square$

## 1.2.F:

- (a) Show that localization commutes with finite products (equivalently, with finite direct sums).

- (b) Show that localization commutes with arbitrary direct sums.
- (c) Show that localization does not necessarily commute with infinite products: the obvious map  $S^{-1}(\prod_i M_i) \rightarrow \prod_i S^{-1}M_i$  induced by the universal property is not always an isomorphism.  
(Hint:  $(1, \frac{1}{2}, \frac{1}{3}, \dots) \in \mathbb{Q} \times \mathbb{Q} \times \mathbb{Q} \times \dots$ )

*Proof.* (a). We need to show there is a bijective  $A$ -module morphism

$$f : S^{-1}(M_1 \times \dots \times M_n) \rightarrow S^{-1}M_1 \times \dots \times S^{-1}M_n$$

Since  $S^{-1}(M_1 \times \dots \times M_n)$  is a localization, the map

$$\phi : M_1 \times \dots \times M_n \rightarrow S^{-1}(M_1 \times \dots \times M_n)$$

is initial among  $A$ -module maps out of  $M_1 \times \dots \times M_n$  where  $s \in S$  acts invertibly. Consider the map

$$\psi : M_1 \times \dots \times M_n \rightarrow S^{-1}M_1 \times \dots \times S^{-1}M_n$$

$$\psi : (m_1, \dots, m_n) \mapsto (\psi_1(m_1), \dots, \psi_n(m_n))$$

Where  $\psi_i : M_i \rightarrow S^{-1}M_i$  is the canonical localization map. This is an  $A$ -module homomorphism out of  $M_1 \times \dots \times M_n$ . And since  $\mu_s \in \text{End}_{\mathbb{Z}}(S^{-1}M_1 \times \dots \times S^{-1}M_n)$  acts componentwise, and  $s$  acts invertibly on each component  $\phi_i(m_i)$ ,  $s$  acts invertibly on the product. So  $\mu_s^{-1} \in \text{End}_{\mathbb{Z}}(S^{-1}M_1 \times \dots \times S^{-1}M_n)$ . Therefore the pair  $(S^{-1}M_1 \times \dots \times S^{-1}M_n, \psi)$  is an object of this category. So by the universal property of localization:

$$\begin{array}{ccc} M_1 \times \dots \times M_n & \xrightarrow{\phi} & S^{-1}(M_1 \times \dots \times M_n) \\ \psi \downarrow & \swarrow \text{dashed } f & \\ S^{-1}M_1 \times \dots \times S^{-1}M_n & & \end{array}$$

$$\exists! f : S^{-1}(M_1 \times \dots \times M_n) \rightarrow S^{-1}M_1 \times \dots \times S^{-1}M_n$$

$$f : \frac{(m_1, \dots, m_n)}{s} \mapsto \left( \frac{m_1}{s}, \dots, \frac{m_n}{s} \right)$$

Which makes the diagram commute. We want to show  $f$  is a bijective  $A$ -module homomorphism.

Surjective: Let  $\left(\frac{m_1}{s_1}, \dots, \frac{m_n}{s_n}\right) \in S^{-1}M_1 \times \dots \times S^{-1}M_n$ . Define  $t = \prod_i s_i$ , since  $S$  is multiplicative,  $t \in S$ . Let  $x_i = \prod_{j \neq i} s_j \in S$ , then

$$\left(\frac{m_1}{s_1}, \dots, \frac{m_n}{s_n}\right) = \left(\frac{x_1 m_1}{t}, \dots, \frac{x_n m_n}{t}\right) = \theta\left(\frac{(x_1 m_1, \dots, x_n m_n)}{t}\right)$$

So  $\theta$  is surjective.

Injective: Let  $\theta\left(\frac{(m_1, \dots, m_n)}{s}\right) = \theta\left(\frac{(m'_1, \dots, m'_n)}{t}\right)$ . So

$$\left(\frac{m_1}{s}, \dots, \frac{m_n}{s}\right) = \left(\frac{m'_1}{t}, \dots, \frac{m'_n}{t}\right)$$

Then componentwise,  $\frac{m_i}{s} = \frac{m'_i}{t} \implies \exists u_i \in S$  such that

$$u_i(m_i t - m'_i s) = 0$$

Then let  $u = \prod_i u_i$ . Clearly  $u \in S$ , and

$$u(m_i t - m'_i s) = 0$$

For all  $i \in I$  simultaneously. So

$$u[(m_1, \dots, m_n)t - (m'_1, \dots, m'_n)s] = 0$$

Therefore

$$\frac{(m_1, \dots, m_n)}{s} = \frac{(m'_1, \dots, m'_n)}{t}$$

As required.

Alternatively: Let  $\theta\left(\frac{(m_1, \dots, m_n)}{s}\right) = 0$ . Then  $\frac{m_i}{s} = 0$  for all  $i$ . Therefore

$$\frac{(m_1, \dots, m_n)}{s} = 0$$

so  $\ker(\theta) = \{0\}$ , and so  $\theta$  is injective.

- (b). Let  $\theta : S^{-1}(\bigoplus_{i \in I} m_i) \rightarrow \bigoplus_{i \in I} S^{-1}M_i$  be the unique  $A$ -module homomorphism given by the universal property of localization, which maps

$$\theta : \frac{(m_i)_{i \in I}}{s} \mapsto \left( \frac{m_i}{s} \right)_{i \in I}$$

It remains to show that  $\theta$  is bijective.

Surjective: Let

$$x = (x_i)_{i \in I} \in \bigoplus_{i \in I} S^{-1}M_i$$

be an arbitrary element. Since it is an element of a direct sum, it has finite support. Let  $J \subset I$  be the (finite) subset of  $I$  such that  $x_j = \frac{m_j}{s_j} \neq 0$  when  $j \in J$ , and  $x_i = 0$  otherwise. The collection of all such  $x_j$  terms defines an isomorphism into  $S^{-1}M_1 \times \cdots \times S^{-1}M_n$ . Then, we use the argument from (a) to see  $\theta$  is surjective.

Injective: Let

$$\theta \left( \frac{(m_i)_{i \in I}}{s} \right) = 0$$

Then  $\frac{m_i}{s} = 0$  for all  $i \in I$ . Let  $J \subset I$  again be the finite subset for which  $m_j \neq 0$ , and  $m_i = 0$  otherwise. If  $j \in J$ , then

$$m_j = 0 \implies \exists u_j \in S \text{ such that } u_j m_j = 0$$

Define  $u = \prod_{j \in J} u_j$ . Then for all  $i$ ,  $u m_i = 0$ . So we have found a  $u \in S$  such that  $u m_i = 0$  for all  $i \in I$ , therefore  $\frac{(m_i)_{i \in I}}{s} = 0$ , and so  $\ker(\theta) = \{0\}$  and  $\theta$  is injective.

- (c). Let  $A = \mathbb{Z}$ ,  $S = \mathbb{Z} \setminus \{0\}$ . Then  $S^{-1}\mathbb{Z} = \mathbb{Q}$ . Consider the element

$$\left( 1, \frac{1}{2}, \frac{1}{3}, \dots \right) \in \mathbb{Q} \times \mathbb{Q} \times \dots$$

A valid preimage of this element would have to contain every positive integer (in particular every prime number) as a factor of some  $s \in S$ . However that is a contradiction because no integer contains every prime as a factor. So

$$\theta : S^{-1}(\prod \mathbb{Z}) \rightarrow \prod \mathbb{Q}$$

Is not an isomorphism.

□

## 1.2.G

Show that

$$\mathbb{Z}/(10) \otimes_{\mathbb{Z}} \mathbb{Z}/(12) \cong \mathbb{Z}/(2)$$

**Remark.** *Note: Over  $\mathbb{Z}$ , the tensor product  $M \otimes_{\mathbb{Z}} N = \{m \otimes n \mid m \in M, n \in N\}$  is completely determined by the pure tensor  $1 \otimes 1$ , because for any  $m \otimes n \in M \otimes_{\mathbb{Z}} N$ , we have*

$$m \otimes n = m(1 \otimes n) = mn(1 \otimes 1)$$

*Proof.* Let  $x \otimes y \in \mathbb{Z}/(10) \otimes_{\mathbb{Z}} \mathbb{Z}/(12)$ . Then,  $x \otimes y = xy(1 \otimes 1)$ . By Bezout, since  $\gcd(10, 12) = 2$ , we can write  $2 = 10(-1) + 12(1)$ . So for even integers  $2k$  we have

$$2k = [10(-1) + 12(1)] k$$

So

$$\begin{aligned} 2k(1 \otimes 1) &= [10(-1) + 12(1)] k(1 \otimes 1) \\ &= -10k(1 \otimes 1) + 12k(1 \otimes 1) \\ &= 0 + 0 = 0 \end{aligned}$$

For odd integers  $2k + 1$  we have

$$2k + 1 = [10(-1) + 12(1)] k + 1$$

So

$$2k + 1(1 \otimes 1) = 2k(1 \otimes 1) + 1(1 \otimes 1) = 1 \otimes 1$$

So there are 2 cases:

- If  $2 \mid xy$ , then  $xy(1 \otimes 1) = 0$
- If  $2 \nmid xy$ , then  $xy(1 \otimes 1) = 1 \otimes 1$

(Furthermore,  $(1 \otimes 1) + (1 \otimes 1) = 2(1 \otimes 1) = 0$ )

Therefore,  $\mathbb{Z}/(10) \otimes_{\mathbb{Z}} \mathbb{Z}/(12) \cong \mathbb{Z}/(2)$ . □

## 1.2.H